

Table of Contents

SOLUTIONS & EXPLANATIONS

Adaptive Digital SAT Practice Test — Horizon Assessment Series — Form A

This booklet contains:

- Quick-reference answer grids for all 6 modules
- Detailed step-by-step explanations for every question
- Why each correct answer is right and why each wrong answer fails

Total: 147 questions (81 Reading & Writing + 66 Math)

These solutions correspond to 100% original practice questions. They are not affiliated with, endorsed by, or sourced from the College Board, ETS, or any official SAT materials.

QUICK-REFERENCE ANSWER GRIDS

Section 1 — Module 1 (27 Qs)

1. D	2. B	3. A	4. A	5. C	6. B	7. D	8. C	9. B
10. A	11. C	12. A	13. C	14. D	15. A	16. B	17. D	18. D
19. A	20. C	21. D	22. B	23. A	24. B	25. C	26. B	27. D

Section 1 — Module 2 Easier (27 Qs)

1. C	2. D	3. B	4. C	5. A	6. B	7. C	8. B	9. A
10. D	11. B	12. A	13. C	14. D	15. B	16. D	17. D	18. C
19. B	20. C	21. D	22. D	23. A	24. A	25. A	26. A	27. B

Section 1 — Module 2 Harder (27 Qs)

1. D	2. A	3. B	4. D	5. A	6. C	7. B	8. A	9. C
10. D	11. B	12. B	13. C	14. C	15. D	16. D	17. D	18. B
19. B	20. C	21. A	22. D	23. A	24. C	25. A	26. A	27. B

Section 2 — Module 1 (22 Qs)

1. D	2. A	3. A	4. 9	5. D	6. A	7. C	8. D	9. 13
10. C	11. B	12. B	13. B	14. 44	15. 300	16. C	17. 42	18. C

1. D	2. A	3. A	4. 9	5. D	6. A	7. C	8. D	9. 13
19. A	20. C	21. D	22. B					

Section 2 — Module 2 Easier (22 Qs)

1. A	2. 3250	3. B	4. A	5. A	6. C	7. C	8. D	9. 22.4
10. B	11. C	12. B	13. D	14. C	15. A	16. B	17. D	18. D
19. C	20. 80	21. -18	22. 2187					

Section 2 — Module 2 Harder (22 Qs)

1. A	2. 75	3. A	4. D	5. 250	6. C	7. C	8. D	9. A
10. 55	11. A	12. B	13. 66	14. -36	15. 4096	16. B	17. D	18. C
19. B	20. B	21. 7	22. C					

Section 1 — Module 1: Detailed Explanations

Question 1 — Answer: D

Vocabulary in Context | Medium

Why correct: The passage contrasts Hicks’s vibrant works with others that have “subdued hues.” The viewer can easily pick out or *discern* those muted tones — meaning to recognize or distinguish them visually.

- A) “Appreciate” implies a value judgment rather than visual recognition.
- B) “Overlook” means to fail to notice — the opposite of the intended meaning.
- C) “Fabricate” means to construct or invent, unrelated to perceiving colors.

Question 2 — Answer: B

Vocabulary in Context | Medium

Why correct: Broadcasters tried to influence how the public viewed or regarded radio. “Perception of” captures this idea of shaping opinion about the medium.

- A) “Contribution to” implies the public added something to radio, reversing the direction.
- C) “Reproduction of” implies copying, not opinion-forming.
- D) “Application to” implies practical use rather than attitude.

Question 3 — Answer: A

Vocabulary in Context | Medium

Why correct: The passage emphasizes thoroughness — consulting entomologists and ecologists extensively. *Rigor* means exactness and carefulness of preparation.

- B) “Obscurity” means being unknown or hard to understand.
 - C) “Shallowness” is the opposite of the deep preparation described.
 - D) “Novelty” means newness, which doesn’t capture thoroughness.
-

Question 4 — Answer: A

Vocabulary in Context | Hard

Why correct: The blank describes firms gathering or concentrating in a region. *Amass in* means to accumulate or collect in a place, fitting the cluster concept.

- B) “Appeal to” means to attract — but firms aren’t being attracted to a quality; they’re physically concentrating.
 - C) “Intercede in” means to intervene on behalf of someone.
 - D) “Concur with” means to agree, which doesn’t describe physical clustering.
-

Question 5 — Answer: C

Vocabulary in Context | Medium

Why correct: Plain acoustic settings “were likely _____ audiences in much earlier periods” because elaborate stages “were not widespread until the 1800s.” Since unadorned stages were the norm, they were *expected by* audiences.

- A) “Disliked by” is too negative — the text implies normalcy, not displeasure.
 - B) “Confusing to” implies bewilderment, which is unsupported.
 - D) “Exciting to” contradicts the idea that plain stages were mundane.
-

Question 6 — Answer: B

Central Ideas and Purpose | Medium

Why correct: The text describes how the Heidelberg Project transforms abandoned lots into art while positively affecting the surrounding community (reducing dumping and arson).

- A) The text doesn’t compare this project to “traditional urban renewal programs.”
 - C) Only some neighbors objected; the text doesn’t claim residents “historically opposed” public art.
 - D) No comparison with other programs across the US is made.
-

Question 7 — Answer: D

Text Structure | Hard

Why correct: The text first presents the favorable opinion of Mrs. Brympton held by servants and neighbors, then describes the specific behaviors (never initiating disagreement, always inquiring politely) that enable her to maintain that favorable reputation.

- A) There is no shift from low to high esteem — she is regarded well throughout.
 - B) No private indifference or suspicion is mentioned.
 - C) No comparison between neighbors' and servants' judgment is drawn.
-

Question 8 — Answer: C

Text Structure | Hard

Why correct: The text describes bioluminescence (a phenomenon), discusses two studies of it (Martinez et al. and Oba et al.), then introduces a third study (Kondo and Fujimoto) that synthesizes both approaches.

- A) The shared-ancestral-gene assumption isn't shown to be "unfounded" — just incomplete.
 - B) The text doesn't critique any study's approach.
 - D) No speculation about future studies appears.
-

Question 9 — Answer: B

Central Ideas | Medium

Why correct: The narrator states being "so keenly conscious of standing at a threshold," explicitly recognizing this as a significant turning point between past obligations and future possibility.

- A) No business opportunity is mentioned.
 - C) The narrator describes stumbling more than succeeding, contradicting a "systematic approach."
 - D) No gratitude toward mentors is expressed.
-

Question 10 — Answer: A

Inferences | Hard

Why correct: The productivity hypothesis holds that groundwater replaces surface water as farm yields rise. An advocate would explain India's increased groundwater use by citing rising farm yields between 1990 and 2015.

- B) Technology access is a different explanation, not the productivity hypothesis.

- C) Rainfall variability is the counter-argument presented by Nair and Subramanian.
 - D) This says the link weakened, which opposes the hypothesis.
-

Question 11 — Answer: C

Command of Quantitative Evidence | Easy

Why correct: The table shows Chile extracted 4.60 billion kg in 2000 and 5.70 billion kg in 2022. Only option C states the correct 2022 value for Chile.

- A) 3.45 is Brazil's 2022 figure.
 - B) 1.80 is Indonesia's 2022 figure.
 - D) 3.20 is Australia's 2022 figure.
-

Question 12 — Answer: A

Textual Evidence (Poetry) | Hard

Why correct: Option A presents a quotation where the speaker describes landscape features that “whisper still of former days,” directly illustrating the claim that winter landscapes carry traces of the past.

- B) This describes general reflection on a calm scene, not traces of the past specifically.
 - C) This describes admiration of ice formations, not the land carrying memory.
 - D) This describes travellers observing frozen streams — external observation, not the landscape encoding the past.
-

Question 13 — Answer: C

Command of Quantitative Evidence (Graph) | Medium

Why correct: The passage says the gap between familiar and unfamiliar satisfaction varied across compositions and asks for the pair showing this variation most clearly. “March” and “Waltz” have the largest gap (~3 points), best illustrating the magnitude of the difference.

- A) “Nocturne” and “Fugue” show moderate gaps, not the most illustrative pair.
 - B) “Rondo” and “Prelude” have nearly equal bars (small difference) — they show little variation.
 - D) “Fugue” and “Prelude” is a mixed pairing without the clearest contrast.
-

Question 14 — Answer: D

Strengthen / Support Claims | Hard

Why correct: The hypothesis is that size differences are an adaptation to prey availability in the Eastern Atlantic. Finding that prey species in that feeding range are unique and unavailable elsewhere directly supports a prey-driven adaptation.

- A) Proximity to fishing fleets is about human activity, not natural prey.
 - B) Stable vs. declining lengths describe a trend but don't explain adaptation to prey.
 - C) Shallow foraging areas don't directly address the type of prey available.
-

Question 15 — Answer: A

Logical Completion | Hard

Why correct: The passage warns against treating *Americanah* as fully autobiographical. Readers who do so risk imposing unwarranted correspondences between the fiction and the author's life.

- B) The passage already acknowledges Adichie drew on personal material — this answer contradicts the stated concern.
 - C) Readership size is irrelevant to the autobiographical question.
 - D) Borrowing techniques from earlier novelists is not discussed.
-

Question 16 — Answer: B

Boundaries (Punctuation) | Easy

Why correct: The sentence requires no punctuation after “from” because it is a preposition linking “donated” to “the private collections.” The prepositional phrase flows naturally without interruption.

- A) A semicolon after “from” incorrectly separates the preposition from its object.
 - C) A colon is not appropriate after a preposition.
 - D) A comma unnecessarily interrupts the prepositional phrase.
-

Question 17 — Answer: D

Form, Structure, and Sense (Verb Forms) | Easy

Why correct: The subject “museums” requires the base verb “educate” in simple present tense to form a complete, grammatically correct sentence: “Greece’s museums educate visitors...”

- A) “Having educated” is a past participle phrase that cannot serve as a main verb.
 - B) “Educating” is a present participle that creates a sentence fragment.
 - C) “To educate” is an infinitive that doesn't complete the predicate.
-

Question 18 — Answer: D

Form, Structure, and Sense (Pronoun Agreement) | Easy

Why correct: “It” is the correct singular pronoun to refer to the singular antecedent “album” (“Harvest”).

- A) “Those” is plural and lacks a clear singular antecedent.
 - B) “They” is plural and doesn’t agree with the singular “album.”
 - C) “Each” doesn’t function as a subject pronoun referring to a specific album.
-

Question 19 — Answer: A

Form, Structure, and Sense (Verb Tense) | Medium

Why correct: The sentence describes a general, ongoing truth about how land trusts function (“the revenue is shared... the members _____ directly”). Simple present tense “benefit” is correct for timeless statements.

- B) “Had benefited” is past perfect, inappropriate for a general truth.
 - C) “Benefited” is simple past.
 - D) “Were benefiting” is past progressive.
-

Question 20 — Answer: C

Boundaries (Punctuation) | Medium

Why correct: A comma before the participial phrase “revealing how...” correctly integrates it as additional modifying information about what the documentary explores.

- A) No punctuation creates a run-on sentence.
 - B) A semicolon requires a complete independent clause to follow.
 - D) A period creates a sentence fragment beginning with “Revealing.”
-

Question 21 — Answer: D

Transitions | Easy

Why correct: The passage contrasts insulated items (retaining heat) with uninsulated items (losing heat). “On the other hand” is a contrast transition.

- A) “As a result” implies cause-effect.
 - B) “Specifically” introduces a narrowing detail.
 - C) “In conclusion” signals a summary.
-

Question 22 — Answer: B

Transitions | Medium

Why correct: “For example” introduces a specific illustration of the general geological principle just described (older strata exposed when deposition ceases).

- A) “Fittingly” implies something is appropriate, not illustrative.
 - C) “Likewise” signals similarity, but the sentence provides an example, not a parallel.
 - D) “Though” signals contrast, but the example supports the statement.
-

Question 23 — Answer: A

Rhetorical Synthesis | Easy

Why correct: To emphasize the size of Matsumoto, the answer must mention its area. Only A states “978 km².”

- B) Discusses services, not size.
 - C) Mentions services provided, not area.
 - D) Mentions the number of municipalities, not Matsumoto’s size.
-

Question 24 — Answer: B

Rhetorical Synthesis | Medium

Why correct: The goal is to emphasize a difference between the two plants. Option B explicitly contrasts their native/non-native status: one is native to Guam; the other is not.

- A) Mentions only one plant.
 - C) Describes the study’s method, not a plant difference.
 - D) Mentions both plants but doesn’t highlight a difference between them.
-

Question 25 — Answer: C

Rhetorical Synthesis | Medium

Why correct: To emphasize chronological order, the answer must state which came first and which came later. Option C — “The Quaternary Ice Age began long after the Karoo Ice Age” — establishes sequence.

- A) Mentions only the Karoo Ice Age.
 - B) Notes similarity (both were cold), not chronological order.
 - D) Describes the general cycle without specific ordering.
-

Question 26 — Answer: B

Rhetorical Synthesis | Medium

Why correct: The goal is to emphasize a similarity. Option B states that both alloys expand when heated — the shared behavior.

- A) Mentions “two research teams,” but the notes describe one study.
 - C) Emphasizes a difference (Invar expanded less), not a similarity.
 - D) General statement that doesn’t mention both alloys specifically.
-

Question 27 — Answer: D

Rhetorical Synthesis | Medium

Why correct: For an audience already familiar with weed suppression, option D presents the specific findings (wood-chip mulch, 10 cm depth, temperate maritime climate) without re-defining the concept.

- A) Mentions only the publication date, not the findings.
 - B) Defines weed suppression, which is unnecessary for a familiar audience.
 - C) Too vague — doesn’t present specific findings.
-

Section 1 — Module 2 Easier: Detailed Explanations

Question 1 — Answer: C

Vocabulary in Context | Easy

Why correct: The marshes “shift in extent and composition” as conditions change, making them *volatile* (subject to rapid or unpredictable change).

- A) “Static” means unchanging — the direct opposite.
 - B) “Immutable” means unchangeable — also opposite.
 - D) “Permanent” contradicts the shifting described.
-

Question 2 — Answer: D

Vocabulary in Context | Easy

Why correct: The play features “elaborate fictional scenarios involving celestial beings,” which anticipates speculative fiction. “Visionary” means ahead of its time or imaginatively forward-looking.

- A) “Authorized” means officially approved.
- B) “Applicable” means relevant — too generic.

- C) “Sarcastic” describes mockery, not speculative imagination.
-

Question 3 — Answer: B

Vocabulary in Context | Medium

Why correct: The findings challenged established consensus, “forcing them to reconsider fundamental assumptions.” This is *disruptive* — causing disturbance to established ideas.

- A) “Gratifying” implies pleasure, but researchers whose views were overturned wouldn’t find it pleasing.
 - C) “Irrelevant” contradicts the significant impact described.
 - D) “Complementary” implies supporting the earlier view — the opposite of what occurred.
-

Question 4 — Answer: C

Central Ideas and Purpose | Easy

Why correct: The text presents two institutions — the Guadalupe Cultural Arts Center and the National Museum of the American Latino — and describes each institution’s cultural focus.

- A) Collection acquisition methods aren’t discussed.
 - B) Collection/exhibition sizes aren’t compared.
 - D) No historical development leading to founding is traced.
-

Question 5 — Answer: A

Text Structure | Medium

Why correct: The text first describes a shared aesthetic framework (bold geometric patterns) used by Gee’s Bend quilters, then illustrates how individuals introduced variations within that framework (Annie Mae Young’s recycled work-clothing fabric).

- B) The text doesn’t say works are perceived as derivative.
 - C) No historical context or exploration of unconventional themes is discussed.
 - D) No collaboration origin story is given.
-

Question 6 — Answer: B

Cross-Text Connections | Hard

Why correct: Espinoza et al. argue that operating multiple OTEC plants could reduce the thermal gradient over time. They would point out that Tanaka et al.’s assumed gradient may not remain stable if multiple plants operate in the same region.

- A) The response isn’t about comparing with other renewables.

- C) Espinoza doesn't claim OTEC plants can't generate electricity — only that conditions may change.
 - D) Chemical composition of deep water isn't discussed.
-

Question 7 — Answer: C

Inferences | Medium

Why correct: The passage describes Ando's 1960s European exposure to Le Corbusier, then notes the Church of the Light's "dramatic interplays of natural light and shadow." This later work reflects the influence of his international architectural exposure.

- A) No "transitional moment" is explicitly described.
 - B) Nothing indicates Ando abandoned this approach.
 - D) The Church is described as different from, not characteristic of, his earliest austere work.
-

Question 8 — Answer: B

Strengthen / Support Claims | Hard

Why correct: The claim is that seawalls are more negatively associated with invertebrate health than groins. Option B directly compares a seawall-heavy site (lower BCHI) to a groin-heavy site (higher BCHI), isolating the variable.

- A) Compares a mixed-structure site to a low-structure site — doesn't isolate seawalls vs. groins.
 - C) Shows equal-structure sites; can't determine which type is more harmful.
 - D) Compares two seawall-heavy sites to each other, not seawalls to groins.
-

Question 9 — Answer: A

Command of Quantitative Evidence (Graph) | Hard

Why correct: The graph shows snow-ice dropping sharply in 2018 while clear-ice stayed steady. Warm January rains would wash away accumulated snow (reducing snow-ice) without affecting clear-ice formation, explaining the selective decline.

- B) A fungal bloom wouldn't affect ice formation.
 - C) The graph doesn't show both types increasing steadily from 2010–2016.
 - D) If both types were disrupted equally, clear-ice would also drop — but the graph shows only snow-ice declining.
-

Question 10 — Answer: D

Command of Quantitative Evidence | Medium

Why correct: The researcher argues GPS yields significantly longer recorded distances than tag-recapture. The monarch butterfly was tracked via tag-recapture (4,800 km). If tracked by GPS (recording all movements including daily foraging flights), the distance would plausibly be much greater.

- A) This discusses the Arctic tern with a different method but in the wrong direction.
 - B) Saying GPS would yield less contradicts the researcher's claim.
 - C) Tag-recapture wouldn't yield more than GPS — this reverses the argument.
-

Question 11 — Answer: B

Strengthen / Support Claims | Medium

Why correct: The text claims steady power output is essential for industrial monitoring. If unreliable power compromises sensor accuracy, that directly explains why steady output matters.

- A) Standard modules working on furnaces doesn't address the Bi₂Te₃ module's advantage.
 - C) Incompatibility would undermine the claim.
 - D) Restates the conclusion rather than providing independent supporting evidence.
-

Question 12 — Answer: A

Logical Completion | Medium

Why correct: The passage catalogs persistent challenges (traffic, parking, weight limits) with no solution fully working. The logical conclusion: logistics companies are unlikely to overcome these impediments in the near term.

- B) Nothing suggests innovations will increase expectations.
 - C) The text doesn't compare warehouse vs. last-kilometer investment.
 - D) This focuses only on cargo bicycles, while the conclusion should be broader.
-

Question 13 — Answer: C

Logical Completion | Medium

Why correct: Researchers must look elsewhere for earlier specimens because the Erfoud deposits are from a significantly earlier era (Ordovician) than the Strud site.

- A) Both sites are in Africa/Morocco — the issue is geological age, not geography.
- B) Global distribution isn't the stated reason.
- D) Preservation quality isn't mentioned as the deciding factor.

Question 14 — Answer: D

Logical Completion | Hard

Why correct: Materials with more open crystal lattices allow ions to migrate with less thermal energy. Since SDC works at 550°C instead of 800°C, the researchers concluded that SDC likely has a more open crystalline structure, enabling ion transport at lower temperatures.

- A) This discusses anode reactions, not electrolyte conductivity.
 - B) Lattice contraction would hinder ion flow, not help it.
 - C) This describes the outcome without explaining the mechanism.
-

Question 15 — Answer: B

Logical Completion | Hard

Why correct: The model uses a uniform silica value where real values likely vary by depth. Therefore, some discrepancies between the model's predictions and observed data should be expected.

- A) Possible but speculative about future sampling efforts.
 - C) Confuses cause and effect within the model's own structure.
 - D) Makes an unsupported claim about actual vs. modeled values.
-

Question 16 — Answer: D

Boundaries (Punctuation) | Easy

Why correct: "Patterns that" creates a restrictive relative clause correctly modifying "polka-dot patterns" — no punctuation break needed.

- A) A period creates a fragment starting with "That."
 - B) A colon is inappropriate before "that."
 - C) A semicolon requires an independent clause after it.
-

Question 17 — Answer: D

Boundaries (Punctuation) | Easy

Why correct: A period correctly separates two complete independent sentences: "...an international phenomenon." and "In 2019, the French duo..."

- A) No punctuation creates a run-on.
- B) A comma creates a comma splice.
- C) "And in" without proper punctuation doesn't properly join these clauses.

Question 18 — Answer: C

Boundaries (Punctuation) | Medium

Why correct: No punctuation is needed after “languages” because the dash that immediately follows (“—including Kannada and Malayalam...—”) already sets off the parenthetical. Adding punctuation before the dash would be redundant.

- A) A comma before a dash is redundant punctuation.
 - B) A colon before a dash is incorrect.
 - D) A second dash here would create three dashes, confusing the sentence structure.
-

Question 19 — Answer: B

Boundaries (Punctuation) | Medium

Why correct: A period correctly separates two complete independent clauses. The first sentence ends with “periodicity.” The second begins “Halley’s Comet is relatively predictable...”

- A) “periodicity; the” would work grammatically but leaves “while the” in a dependent position.
 - C) “periodicity, while the” creates a comma splice.
 - D) “periodicity: while the” misuses a colon before a dependent clause.
-

Question 20 — Answer: C

Form, Structure, and Sense (Verb Forms) | Medium

Why correct: The subject “mechanism” is singular, and the sentence describes a general ongoing truth, requiring present tense: “counteracts.”

- A) “Had counteracted” is past perfect — inappropriate for a general scientific statement.
 - B) “Counteracted” is simple past.
 - D) “Counteracting” is a participle that would make the sentence a fragment.
-

Question 21 — Answer: D

Form, Structure, and Sense (Verb Tense) | Medium

Why correct: The compound subject “patterns and... data” is plural, requiring the plural verb “determine.”

- A) “Determines” is singular — disagrees with the compound subject.
- B) “Has determined” is singular present perfect.
- C) “Determining” is a participle creating a fragment.

Question 22 — Answer: D

Transitions | Easy

Why correct: “Next” logically signals the second step in a sequential comparison: “first consider X; next, compare them to Y.”

- A) “Therefore” implies a logical conclusion.
 - B) “Still” implies concession or persistence.
 - C) “Instead” implies substitution, not sequence.
-

Question 23 — Answer: A

Transitions | Medium

Why correct: “Accordingly” signals a logical result: because the plant absorbs cadmium from the soil, soil cadmium levels decrease as a consequence.

- B) “Specifically” introduces a detail, not a consequence.
 - C) “Nevertheless” signals contrast.
 - D) “In addition” adds information without showing causation.
-

Question 24 — Answer: A

Rhetorical Synthesis | Easy

Why correct: The goal is to give an example of a country using a cognate of “yuan.” Japan uses “yen” (derived from “yuan”), making “Japan uses the yen” the clearest example.

- B) This negates the connection.
 - C) This explains etymology without naming a country as an example.
 - D) Too general — doesn’t name a specific country.
-

Question 25 — Answer: A

Rhetorical Synthesis | Medium

Why correct: The goal is to present biographical analysis to an unfamiliar audience. Option A defines the concept clearly: “considers the life circumstances of the author when interpreting a work.”

- B) Provides an example without defining the concept first.
 - C) Discusses formalist analysis instead.
 - D) Compares the two approaches without explaining what biographical analysis is.
-

Question 26 — Answer: A

Rhetorical Synthesis | Medium

Why correct: The goal is to emphasize a difference. Option A explicitly contrasts the two: nanowires have “elongated cylinders” while quartz has a “continuous SiO₄ framework.”

- B) Notes both are silicon-based (a similarity, not a difference).
 - C) Incorrectly calls quartz “cylindrical.”
 - D) Inaccurately describes SiO₄ tetrahedra forming tubes in nanowires.
-

Question 27 — Answer: B

Rhetorical Synthesis | Hard

Why correct: The goal is to compare Simon and Gigerenzer. Option B directly contrasts them: Simon sees heuristics as simplifications due to limits; Gigerenzer reframes them as efficient strategies.

- A) Simon didn’t specifically claim cognition is constrained by “neural architecture.”
 - C) Neither framework focuses on perception-level processes.
 - D) Gigerenzer doesn’t simply extend Simon’s concept — he reinterprets heuristics positively.
-

Section 1 — Module 2 Harder: Detailed Explanations

Question 1 — Answer: D

Vocabulary in Context | Hard

Why correct: The oral histories “carry legal and cultural authority in treaty negotiations,” making them *juridical* (relating to law or legal proceedings).

- A) “Provisional” means temporary — contradicts “continue to carry authority.”
 - B) “Ornamental” means decorative — dismisses their substantive legal function.
 - C) “Speculative” means theoretical — but these records are presented as authoritative.
-

Question 2 — Answer: A

Vocabulary in Context | Hard

Why correct: The meta-analysis found fragmentation had “neutral or even positive effects” in nearly half the studies — the relationship is *equivocal* (ambiguous, open to multiple interpretations).

- B) “Negligible” means insignificant — but the relationship was complex, not absent.

- C) “Corrosive” means destructive — doesn’t describe a statistical relationship.
 - D) “Transparent” means clear — the opposite of what’s described.
-

Question 3 — Answer: B

Vocabulary in Context | Medium

Why correct: The exhibition’s influence works through “debates and conceptual experiments” rather than reshaping buildings directly. The influence is therefore *indirect*.

- A) “Tangible” means physically touchable — opposite of indirect influence.
 - C) “Instantaneous” contradicts “over subsequent years.”
 - D) “Diminishing” implies declining influence, which isn’t stated.
-

Question 4 — Answer: D

Central Ideas and Purpose | Hard

Why correct: The text describes Flores’s argument that conventional narratives overlook essential contributions by rural women during the Mexican Revolution.

- A) The text doesn’t rank women’s contributions as “more important” than soldiers’.
 - B) No comparison of military strategies is made.
 - C) Methodology isn’t the focus.
-

Question 5 — Answer: A

Text Structure | Hard

Why correct: The text introduces a widely held view (reading is passive), rejects it (“hopelessly crude”), then supports the rejection with an example (a metaphor on page forty recoloring everything).

- B) Only one view is presented and rejected — no two competing theories.
 - C) No historical change in attitudes is traced.
 - D) No survey of multiple answers is conducted.
-

Question 6 — Answer: C

Cross-Text Connections | Hard

Why correct: Osei et al. found introduced microbes were displaced within weeks. They would argue Wei et al.’s increased root biomass was likely caused by a temporary nutrient boost from the slurry, not lasting microbial colonization.

- A) Osei et al. don’t claim field settings are inherently more effective.

- B) Wei et al. did include untreated controls.
 - D) The nitrogen-fixing bacteria issue isn't about whether they were indigenous.
-

Question 7 — Answer: B

Inferences | Hard

Why correct: Chang shows that wealthy nations used protectionist strategies during development but now prescribe free trade to developing countries. This effectively denies those economies access to the same strategies that today's wealthy nations once employed.

- A) Chang doesn't validate free-trade models — he critiques them.
 - C) Geography isn't Chang's focus.
 - D) Chang's point is about denying developmental strategies, not about specializing in comparative advantages.
-

Question 8 — Answer: A

Strengthen / Support Claims | Hard

Why correct: The hypothesis links *L. rhamnosus* to reduced anxiety via GABA signaling. Mice with the bacterium showing elevated GABA receptor expression AND reduced anxiety directly supports this specific mechanism.

- B) This shows germ-free mice are anxious generally but doesn't link *L. rhamnosus* to GABA specifically.
 - C) Increased serotonin without GABA changes would actually weaken the GABA-specific hypothesis.
 - D) Reduced anxiety from eliminating bacteria would contradict the hypothesis.
-

Question 9 — Answer: C

Command of Quantitative Evidence (Graph) | Hard

Why correct: By 2020, Neighborhood C (~25%) exceeded Neighborhood A (~18%), despite starting at less than half of A's coverage in 2005 (C: ~10% vs. A: ~30%).

- A) Neighborhood C gained canopy, not lost it.
 - B) B's coverage peaked in 2015 then dropped sharply — not consistent throughout.
 - D) By 2020, A (~18%) no longer had the highest coverage — C (~25%) surpassed it.
-

Question 10 — Answer: D

Logical Completion | Hard

Why correct: The model uses a constant value where real values likely vary. Therefore, some disagreements between predictions and observed proxy data should be expected.

- A) Possible but speculative about future sampling.
 - B) Makes an unsupported claim about actual vs. modeled values.
 - C) Confuses internal model variations with the model's fixed assumption.
-

Question 11 — Answer: B

Logical Completion | Hard

Why correct: MOF membranes have precisely sized molecular pores. The logical conclusion: the membrane acts as a selective barrier, letting small lithium ions through while physically blocking larger polysulfide molecules.

- A) Saturation would undermine the benefit, not explain the improvement.
 - C) Catalyzing more polysulfide formation would worsen the problem.
 - D) Li-ion batteries don't experience the polysulfide shuttle effect.
-

Question 12 — Answer: B

Boundaries (Punctuation) | Easy

Why correct: A period correctly separates two independent clauses: "...held deep ritual significance." and "The archaeological evidence..."

- A) "significance, the" creates a comma splice.
 - C) A semicolon would work grammatically but B is the best answer here.
 - D) Missing punctuation creates a run-on sentence.
-

Question 13 — Answer: C

Boundaries (Punctuation) | Medium

Why correct: A comma after "construction" correctly closes the participial phrase "completed after more than two decades of construction," allowing the main clause to continue.

- A) A semicolon requires an independent clause after it.
 - B) A colon implies a list or explanation follows.
 - D) No punctuation makes the sentence difficult to parse.
-

Question 14 — Answer: C

Form, Structure, and Sense (Verb Forms) | Hard

Why correct: Present tense “introduces” correctly describes a general, timeless truth: rate variation introduces systematic bias into estimates.

- A) “Had introduced” is past perfect — inappropriate for a general principle.
 - B) “Introducing” creates a participle/fragment.
 - D) “Introduced” is past tense — not suitable for a timeless scientific statement.
-

Question 15 — Answer: D

Form, Structure, and Sense (Verb Tense) | Hard

Why correct: The compound subject “alignment and coalition history” is plural, requiring the plural verb “determine.”

- A) “Determines” is singular — disagrees with the compound subject.
 - B) “Has determined” is singular present perfect.
 - C) “Determining” creates a fragment.
-

Question 16 — Answer: D

Boundaries (Punctuation) | Hard

Why correct: A comma before “and” correctly joins two independent clauses in a compound sentence: “...share a common structural foundation, and the theory has prompted...”

- A) Starting a sentence with “And” after a period is informal for this register.
 - B) Omitting the comma before “and” is acceptable in some styles but less clear with long independent clauses.
 - C) A semicolon before “and” is redundant.
-

Question 17 — Answer: D

Transitions | Medium

Why correct: “Then” signals the second step in a chronological comparison: “first examine X; then compare it to Y.”

- A) “Therefore” implies a logical conclusion.
 - B) “Still” implies concession.
 - C) “Instead” implies replacement or substitution.
-

Question 18 — Answer: B

Transitions | Hard

Why correct: “Consequently” signals a result: because the fern sequesters arsenic in its fronds, soil arsenic levels decline as a consequence.

- A) “Specifically” introduces a detail, not a consequence.
 - C) “Nevertheless” implies contrast.
 - D) “In addition” adds information without showing causation.
-

Question 19 — Answer: B

Rhetorical Synthesis | Hard

Why correct: The goal is to compare Fricker and Dotson. Option B directly contrasts them: Fricker locates harm in the listener’s prejudiced judgments; Dotson highlights how speakers preemptively self-silence in anticipation of such prejudice.

- A) Dotson doesn’t disagree about epistemic injustice being structural.
 - C) Institutional policies aren’t the focus of either concept.
 - D) Dotson doesn’t claim all forms involve both self-censorship and prejudice.
-

Question 20 — Answer: C

Rhetorical Synthesis | Hard

Why correct: The goal is to emphasize the difference in energy yield. Option C directly states the 18-fold difference: “2 ATP vs. roughly 36 ATP.”

- A) Emphasizes similarity (“both convert glucose”), not the energy difference.
 - B) Describes only fermentation without comparing.
 - D) Emphasizes oxygen requirements, not energy yield.
-

Question 21 — Answer: A

Inferences | Hard

Why correct: Ostrom documented communities successfully self-governing common-pool resources for centuries. Her work implies that the state-vs-privatization dichotomy overlooks the capacity of communities to build effective self-governing institutions.

- B) Contradicts Ostrom’s findings.
 - C) Ostrom studied multiple resource types (fisheries, irrigation, forests), not just two.
 - D) Ostrom’s evidence was consistent across regions.
-

Question 22 — Answer: D

Strengthen / Support Claims | Hard

Why correct: Kennett hypothesizes that drought triggered warfare and collapse. If warfare was already common during periods of above-average rainfall, drought cannot be the primary trigger — this weakens the hypothesis.

- A) Stable institutions despite drought weakens it, but less directly than D.
 - B) Consistent data would support, not weaken.
 - C) Water management is a partial buffer but doesn't directly undermine the drought-warfare causal link.
-

Question 23 — Answer: A

Command of Quantitative Evidence | Hard

Why correct: The graph shows other renewables at ~50 EJ in 2020, surpassing nuclear energy at ~20 EJ in absolute contribution.

- B) Fossil fuels grew from ~450 to ~510 EJ — they didn't decline.
 - C) Hydro grew in absolute terms but its share relative to total energy didn't clearly increase.
 - D) Fossil fuels accounted for the majority of the absolute energy increase.
-

Question 24 — Answer: C

Logical Completion | Hard

Why correct: If each new generation accepts depleted conditions as “normal,” then historical ecological data — even when imprecise — is essential for setting accurate restoration benchmarks that reflect pre-depletion conditions.

- A) Pauly's concept argues the opposite — that shifting baselines apply broadly, not just to marine biology.
 - B) The concept warns against exactly this reasoning — focusing only on recent data.
 - D) The concept suggests that “more achievable” targets may actually be miscalibrated.
-

Question 25 — Answer: A

Boundaries (Punctuation) | Hard

Why correct: A comma correctly separates the two items in an appositional list: “abstract freedom, which exists as a theoretical possibility, and situated freedom, which is constrained by...”

- B) No comma makes the long compound harder to parse.
 - C) A semicolon is too strong for items in a coordinated pair.
 - D) Starting with “And” after a period is informal.
-

Question 26 — Answer: A

Form, Structure, and Sense (Verb Forms) | Hard

Why correct: “Is converted” provides the passive voice present tense needed: “the rate at which thawed carbon is converted to atmospheric CO₂.”

- B) “Converting” creates a fragment.
 - C) “Having converted” implies a completed past action.
 - D) “To convert” is an infinitive that doesn’t grammatically fit the clause structure.
-

Question 27 — Answer: B

Rhetorical Synthesis | Hard

Why correct: The goal is to present evidence supporting the conclusion. Option B connects the findings: *mcr-1* was found in a majority of hospital wastewater species, and its downstream detection in municipal wastewater suggests resistance genes spread from hospitals into the wider environment.

- A) A general statement about HGT, not specific evidence from the study.
 - C) Explains why *mcr-1* matters but doesn’t present the study’s evidence chain.
 - D) Notes detection at both sites without connecting hospital origin to downstream spread.
-

Section 2 — Module 1: Detailed Explanations

Question 1 — Answer: D

Linear Equations | Easy

Width = 6 cm. Length = 6 + 40 = 46 cm. Area = 6 × 46 = **276 cm²**.

- A) 12 = 6 + 6 (perimeter error). B) 46 is just the length, not the area. C) 92 = 2(6 + 40) — this is the perimeter.
-

Question 2 — Answer: A

Linear Equations (Interpretation) | Easy

In $28h + 18f = 924$, the coefficient 28 multiplies h (hours at the publishing house). It represents the hourly rate at the publishing house: **\$28 per hour**.

- B) 18 is the law firm rate. C) h represents the number of hours, not 28. D) f represents law firm hours.
-

Question 3 — Answer: A

Data Analysis (Scatterplot) | Medium

The line passes through approximately (0, 1) with slope ≈ 1.5 . Equation: $y = 1.0 + 1.5x$.

- B) Negative slope contradicts the upward trend. C) y-intercept of -1 doesn't match the graph. D) Both negative intercept and slope contradict the upward-trending graph.
-

Question 4 — Answer: 9

Systems of Equations | Hard

Setting $-0.25 = x^2 + 6x + a$ gives $x^2 + 6x + (a + 0.25) = 0$. No real solutions when discriminant < 0 : $36 - 4(a + 0.25) < 0 \rightarrow 35 < 4a \rightarrow a > 8.75$. The least positive integer is **9**.

Question 5 — Answer: D

Geometry (Area) | Easy

Width = 5 cm. Length = $5 + 35 = 40$ cm. Area = $5 \times 40 = \mathbf{200 \text{ cm}^2}$.

- A) $10 = 2 \times 5$. B) 40 is just the length. C) $80 = 2(5 + 35)$ — the perimeter.
-

Question 6 — Answer: A

Ratios | Easy

Blue: 32. Total: $48 + 32 = 80$. Fraction blue = $32/80 = \mathbf{2/5}$.

- B) $3/5 = 48/80$ is the fraction of red marbles. C) $2/3$ doesn't simplify from these numbers. D) $3/8 = 30/80$, incorrect.
-

Question 7 — Answer: C

Linear Functions | Easy

$g(0) = 5(0) - 7 = -7$. The y-intercept is **(0, -7)**.

- A) (0, -5) confuses the coefficient with the constant. B) (0, 5) is the slope, not the y-intercept. D) (0, 7) gets the sign wrong on the constant.
-

Question 8 — Answer: D

Exponential Functions | Medium

Decreasing by 18% means retaining $82\% = 0.82$ per period of 15.5 hours. The exponent must be $t/15.5$ to scale correctly: $P = 50(0.82)^{(t/15.5)}$.

- A) Uses 0.18 (the decay rate) and wrong exponent form. B) Uses 0.18 instead of 0.82. C) Uses $t + 15.5$ instead of $t/15.5$ in the exponent.
-

Question 9 — Answer: 13

Equivalent Expressions | Medium

$5 + 11 - 3 = 13$. So $5x^4 + 11x^4 - 3x^4 = 13x^4$.

Question 10 — Answer: C

Exponents and Radicals | Medium

$(245y)^{(1/2)} = \sqrt{(245y)}$. The exponent $1/2$ applies to the entire product inside the parentheses.

- A) $245\sqrt{y}$ incorrectly separates 245 from the square root. B) $\sqrt{245} \cdot y$ takes the root of only 245 and leaves y unsquared. D) $\sqrt{((245y)^2)} = 245y$ — this squares then roots, yielding the original product.
-

Question 11 — Answer: B

Probability | Medium

$P(Z) = 1 - 0.60 - 0.28 = 0.12$. Number in Hall Z = $0.12 \times 350 = 42$.

- A) 12 is a computation error. C) $56 = 0.16 \times 350$. D) $98 = 0.28 \times 350$, which is Hall Y's count.
-

Question 12 — Answer: B

Linear Equations (Perpendicular) | Medium

Perpendicular slopes are negative reciprocals. Slope of $m = -1/3$, so slope of $n = 3$.

- A) 10 is the y -intercept. C) $1/3$ is the absolute value of m 's slope, not the negative reciprocal. D) $1/10$ confuses the y -intercept with slope.
-

Question 13 — Answer: B

Exponential Functions (Tables) | Hard

$f(x) = -8(3)^{(x/3)}$. Evaluating: $f(-3) = -8(3)^{(-1)} = -8/3$. $f(0) = -8(1) = -8$. $f(3) = -8(3) = -24$. $f(6) = -8(9) = -72$. Only option B matches all four values.

- A) $f(0) = 0$ is wrong; $3^0 = 1$, so $f(0) = -8$. C) $f(-3) = 8/3$ has the wrong sign. D) $f(6) = -48$ uses the wrong power of 3.
-

Question 14 — Answer: 44

Function Evaluation | Medium

$$r(2) = 4(2) + 7 = 15. \quad q(4) = 5 - 4 = 1. \quad 3r(2) - q(4) = 3(15) - 1 = 45 - 1 = \mathbf{44}.$$

Question 15 — Answer: 300

Linear Inequalities | Medium

Cost per laptop with 8% tax: $210 \times 1.08 = 226.80$. Max laptops: $68,040 \div 226.80 = \mathbf{300}$ exactly.

Question 16 — Answer: C

Trigonometry | Medium

To convert radians to degrees, multiply by $180^\circ/\pi$. The expression $(7/10)\pi \cdot (180^\circ/\pi)$ correctly converts the radian measure to degrees, yielding $\mathbf{126^\circ}$.

- A) Omits π from the original radian measure. B) $\pi/180^\circ$ converts degrees to radians — the reverse direction. D) $360^\circ/\pi$ is the wrong conversion factor.
-

Question 17 — Answer: 42

Geometry (Parallel Lines) | Hard

Triangle DEF: $96^\circ + (2x+6)^\circ + (3x-12)^\circ = 180^\circ \rightarrow 5x + 90 = 180 \rightarrow x = 18$. So $E = 42^\circ$, $F = 42^\circ$. Since $GH \parallel DF$, by corresponding angles (transversal EF), angle GHE = angle DFE = $\mathbf{42^\circ}$.

Question 18 — Answer: C

Quadratic Functions | Hard

$f(x) = a(x-2)^2 + 5$. Using point (3, 41): $a(1) + 5 = 41 \rightarrow a = 36$. Verify: $f(0) = 36(4) + 5 = 149$ ✓. $f(-1) = 36(9) + 5 = 329$. $f(-1) - f(0) = 329 - 149 = \mathbf{180}$.

- A) 108 is a computation error. B) 252 uses an incorrect coefficient. D) 288 misapplies the vertex form.
-

Question 19 — Answer: A

Data Analysis (Scatterplot) | Medium

The line passes through (0, 2) with slope ≈ 2 : $y = 2 + 2x$.

- B) Negative slope contradicts the upward trend. C) y-intercept of -2 doesn't match. D) This shows a negative slope from $y = 22$.
-

Question 20 — Answer: C

Systems of Linear Equations | Medium

Adding both equations: $(x+4) + (x-4) = (3y+10) + (-3y+2) - 2x = 12 - x = 6$. Therefore $3x = 18$.

- A) $-15/7$ is from an incorrect setup. B) 9 is half the correct answer. D) 36 doubles the correct answer.
-

Question 21 — Answer: D

Exponential Functions (Decay) | Hard

Decreasing by 30% means retaining 70% = 0.70 per period of 18.4 hours: $M = 200(0.70)^{(t/18.4)}$.

- A) Uses 0.30 and wrong exponent form. B) Uses 0.30 (the loss rate) instead of 0.70. C) Uses $t + 18.4$ instead of $t/18.4$.
-

Question 22 — Answer: B

Quadratic Functions (Vertex Form) | Hard

$g(x) = a(x-3)^2 + 10$. Using (4, 46): $a(1) + 10 = 46 \rightarrow a = 36$. Verify: $g(1) = 36(4) + 10 = 154$ ✓. $g(-2) = 36(25) + 10 = 910$. $g(0) = 36(9) + 10 = 334$. $g(-2) - g(0) = 910 - 334 = 576$.

- A) 432 uses an incorrect coefficient. C) 720 is a computation error. D) 864 misapplies the vertex form.
-

Section 2 — Module 2 Easier: Detailed Explanations

Question 1 — Answer: A

Linear Equations (Interpretation) | Easy

The coefficient 15 multiplies n (neighborhood walks), so 15 represents the **earnings per neighborhood walk**.

- B) 12 is the park walk rate. C) n represents the count, not 15. D) p represents park walk count.
-

Question 2 — Answer: 3250

Linear Equations | Easy

Weight = $2,400 + 85(10) = 2,400 + 850 = \mathbf{3,250 \text{ grams}}$.

Question 3 — Answer: B

Linear Equations | Easy

$m - 27n = p \rightarrow \mathbf{m = p + 27n}$. Add $27n$ to both sides.

- A) Subtracts $27n$ instead of adding. C) Multiplies instead of adding. D) Divides instead of adding.
-

Question 4 — Answer: A

Linear Inequalities | Medium

The shaded region is below and left of the line. Point $(-3, 0)$ falls in this region. Checking: when $x = -3$, the line's y-value is positive (above 0), so $(-3, 0)$ is below the line — in the shaded region.

- B) $(0, 8)$ is above the y-intercept of ~ 7 — outside the shaded region. C) $(0, -5)$ may not be in the region. D) $(5, 0)$ is to the right of the line — outside the shaded region.
-

Question 5 — Answer: A

Quadratic Functions | Easy

The function $y = (36w)(w)$ shows the area. The factor $36w$ represents the **length** (36 times the width w).

- B) y is the area, not $36w$. C) The difference would be $36w - w = 35w$, not $36w$. D) w is the width; $36w$ is the length.
-

Question 6 — Answer: C

Linear Equations | Easy

$7(x+3) - 6(x+3) = 28 - (x+3)(7-6) = 28 - \mathbf{x + 3 = 28}$.

- A) -4 would require $x + 3 = -4$, which doesn't satisfy the equation. B) 25 confuses x with $x + 3$. D) 31 adds 3 to 28 unnecessarily.
-

Question 7 — Answer: C

Systems of Equations | Medium

Adding: $(x+8) + (x-8) = (-5y+3) + (5y+19) \rightarrow 2x = 22$. So **$2x = 22$** .

- A) $-7/5$ is a different quantity. B) 11 is x , not $2x$. D) 44 is $4x$, not $2x$.
-

Question 8 — Answer: D

Linear Equations | Easy

The squirrel gathered a total of 52.5 acorns: **$y = 52.5$** .

- A) 10 is the number of days. B) $52.5/10 = 5.25$ is the daily rate, not the total. C) $52.5 + 10 = 62.5$ incorrectly adds days to total.
-

Question 9 — Answer: 22.4

Geometry (Volume) | Medium

Volume = $4.000^3 = 64.000 \text{ cm}^3$. Mass = density $\times$ volume = $0.350 \times 64 = \mathbf{22.4 \text{ grams}}$.

Question 10 — Answer: B

Absolute Value | Medium

$-3|7x+4| + 6 = -6 \rightarrow -3|7x+4| = -12 \rightarrow |7x+4| = 4$. Case 1: $7x+4 = 4 \rightarrow x = 0$. Case 2: $7x+4 = -4 \rightarrow x = -8/7$. Solutions: **0 and $-8/7$** .

- A) Only 0 — misses the second solution. C) $-15/7$ doesn't satisfy the equation. D) Two valid solutions exist.
-

Question 11 — Answer: C

Linear Functions | Medium

$h(x) = 9x + b$. $h(3) = 27 + b = 32 \rightarrow b = 5$. $h(d) = 9d + 5 = -4 \rightarrow d = -1$. $u(-1) = -4$ and $u(6) = 52$. Slope = $(52 - (-4))/(6 - (-1)) = 56/7 = \mathbf{8}$.

- A) -2 is from a sign error. B) 7 is the denominator, not the slope. D) 9 is h 's slope, not u 's.
-

Question 12 — Answer: B

Linear Equations (Graph) | Medium

The line passes through (0, 8) and has positive slope ~ 6 . For the form $Rx - 6y = -48$: at (0, 8): $R(0) - 6(8) = -48 \rightarrow -48 = -48$ ✓. The equation **$Rx - 6y = -48$** fits.

- A) $Rx + 6y = 48$ gives $48 = 48$ at (0, 8) ✓, but the slope sign is wrong. C) $6x + Ry = 48$ doesn't match the y-intercept. D) $6x - Ry = -48$ produces the wrong slope direction.
-

Question 13 — Answer: D

Linear Inequalities | Hard

The line passes through (-9, 0) and (0, -12). Substituting into $rx + ty = -108$: at (-9, 0): $-9r = -108 \rightarrow r = 12$. At (0, -12): $-12t = -108 \rightarrow t = 9$. So $r + t = 12 + 9 = \mathbf{21}$.

- A) -3 comes from a sign error. B) 3 mistakes the relationship between r and t. C) -21 reverses the sign.
-

Question 14 — Answer: C

Geometry (Surface Area) | Hard

Edge ratio = $5/72$. Surface area ratio = $(\text{edge ratio})^2 = (5/72)^2 = \mathbf{25/5184}$.

- A) $125/373248$ is the volume ratio. B) $5/432$ uses an incorrect calculation. D) $25/72$ forgets to square the denominator.
-

Question 15 — Answer: A

Exponential Functions (Graph) | Hard

The graph of $y = g(x) + 3$ passes through (0, 3) and approaches $y = 4$ as $x \rightarrow -\infty$. So $g(0) + 3 = 3 \rightarrow g(0) = 0$, and $g(x) + 3 \rightarrow 4 \rightarrow g(x) \rightarrow 1$. Testing $g(x) = -4^x + 1$: $g(0) = -1 + 1 = 0$ ✓. As $x \rightarrow -\infty$: $g(x) \rightarrow 0 + 1 = 1$ ✓.

- B) $g(0) = -1 + 3 = 2$, but we need $g(0) = 0$. C) $g(0) = -1 + 4 = 3$, but we need $g(0) = 0$. D) $g(0) = -1 + 5 = 4$, but we need $g(0) = 0$.
-

Question 16 — Answer: B

Equivalent Expressions | Easy

$12(x + 8) = 12x + 12(8) = \mathbf{12x + 96}$.

- A) Forgot to distribute 12 to 8. C) Added instead of multiplying: $12 + 8 = 20$. D) Ignored the 8 entirely.

Question 17 — Answer: D

Ratios | Medium

Original: $40 \text{ in} \times 16 \text{ mi/in} = 640 \text{ mi}$ total. Smaller: $40 - 60\%(40) = 16 \text{ in}$ side. Same area covered: $640 \text{ mi} \div 16 \text{ in} = \mathbf{40 \text{ mi/in}}$.

- A) $6.40 = 16 \times 0.40$. B) $9.60 = 16 \times 0.60$. C) 25.60 applies the 60% reduction to the scale instead of the side length.
-

Question 18 — Answer: D

Systems (Quadratic-Linear) | Hard

Substituting $y = x - d$ into $y = -(x-3)^2$: $x - d = -(x-3)^2$. Rearranging: $(x-3)^2 + x - d = 0 \rightarrow x^2 - 5x + (9-d) = 0$. Discriminant: $25 - 4(9-d) = -11 + 4d$. Two solutions when discriminant > 0 : $d > 11/4 = 2.75$. Only $\mathbf{d = 4}$ satisfies $d > 2.75$.

- A) $d = 1 < 2.75$ — no real solutions. B) $d = 2 < 2.75$ — no real solutions. C) $d = 11/4 = 2.75$ — exactly one solution (boundary), not two distinct.
-

Question 19 — Answer: C

Geometry (Similarity) | Hard

For AA similarity, we need two pairs of equal angles. Given: $G = J = 72^\circ$. Option C: $H = 40^\circ \rightarrow I = 180 - 72 - 40 = 68^\circ$. $L = 68^\circ \rightarrow K = 180 - 72 - 68 = 40^\circ$. Both triangles have angles $\{72^\circ, 40^\circ, 68^\circ\}$.

Similar by AA.

- A) Same side lengths ($30 = 30$) don't prove similarity when other sides differ. B) Side ratios GH/GI vs. KL/JL don't establish SAS similarity without the included angle. D) $H = 72^\circ$, $K = 50^\circ$: GHI has $\{72^\circ, 72^\circ, 36^\circ\}$; JKL has $\{72^\circ, 50^\circ, 58^\circ\}$ — different angle sets.
-

Question 20 — Answer: 80

Geometry (Circumscribed Circle) | Hard

Side $= 720/3 = 240 \text{ cm}$. Circumradius $R = a/\sqrt{3} = 240/\sqrt{3} = 240\sqrt{3}/3 = 80\sqrt{3}$. So $\mathbf{w = 80}$.

Question 21 — Answer: -18

Functions (Evaluation) | Hard

$g(x) = |x|/a + 18$ where $a < 0$. $g(20a) = |20a|/a + 18 = 20|a|/a + 18$. Since $a < 0$, $|a| = -a$, so $20(-a)/a = -20$. $g(20a) = -20 + 18 = -2$. $g(9a) = |9a|/a + 18 = -9 + 18 = 9$. Product: $(-2)(9) = -18$.

Question 22 — Answer: 2187

Exponential Functions | Hard

$h(x) = ab^{(x/n)}$. $h(7)/h(4) = b^{(3/n)} = 243/9 = 27 = 3^3$. So $b^{(1/n)} = 3$. $h(4) = a3^4 = 81a = 9 \rightarrow a = 1/9$. $h(9) = (1/9)3^9 = 19683/9 = \mathbf{2187}$.

Section 2 — Module 2 Harder: Detailed Explanations

Question 1 — Answer: A

Linear Equations | Easy

In $C = 250 + 18g$, the constant 250 is the **flat fee** charged regardless of the number of guests.

- B) 18 is the cost per guest. C) 250 isn't a guest count — it's the base fee. D) The equation doesn't solve for when $C = 250$ in terms of guests.
-

Question 2 — Answer: 75

Linear Equations | Medium

Depth = $12.5 \times 6 = \mathbf{75 \text{ meters}}$ after 6 minutes.

Question 3 — Answer: A

Linear Equations | Easy

$p - 42r = s \rightarrow \mathbf{p = s + 42r}$. Add $42r$ to both sides.

- B) Subtracts instead of adding. C) Multiplies instead of adding. D) Divides instead of adding.
-

Question 4 — Answer: D

Systems (Quadratic-Linear) | Hard

Substituting: $x - k = -2(x-5)^2$. Rearranging: $2(x-5)^2 + x - k = 0 \rightarrow 2x^2 - 19x + (50-k) = 0$. Discriminant: $361 - 8(50-k) = -39 + 8k$. Two solutions when $k > 39/8 = 4.875$. Only **$k = 6$** satisfies $k > 4.875$.

- A) $k = 3 < 4.875$ — no solutions. B) $k = 4 < 4.875$ — no solutions. C) $k = 39/8 = 4.875$ — boundary (exactly one solution).
-

Question 5 — Answer: 250

Linear Inequalities | Medium

Cost per chair with 6% tax: $185 \times 1.06 = \$196.10$. Max chairs: $49,210 \div 196.10 \approx 250.94$. Since you can only buy whole chairs, max = **250**.

Question 6 — Answer: C

Absolute Value | Hard

$-2|9x+6| + 8 = -4 \rightarrow |9x+6| = 6$. Case 1: $9x+6 = 6 \rightarrow x = 0$. Case 2: $9x+6 = -6 \rightarrow x = -12/9$. Solutions: **0 and $-12/9$** .

- A) Only 0 — misses the second solution. B) $-2/3$ doesn't satisfy the equation. D) Two valid solutions exist.
-

Question 7 — Answer: C

Linear Functions | Hard

$v(x) = 8x + b$. $v(2) = 16 + b = 23 \rightarrow b = 7$. $v(c) = 8c + 7 = -1 \rightarrow c = -1$. $w(-1) = -1$, $w(7) = 71$. Slope $= (71 - (-1)) / (7 - (-1)) = 72/8 = \mathbf{9}$.

- A) -2 is a sign error. B) 8 is v 's slope, not w 's. D) 12 is from a different calculation.
-

Question 8 — Answer: D

Exponential Functions (Graph) | Hard

The graph of $y = f(x) + 2$ passes through $(0, 8)$ and approaches $y = 9$. So $f(0) + 2 = 8 \rightarrow f(0) = 6$, and $f(x) + 2 \rightarrow 9 \rightarrow f(x) \rightarrow 7$. Testing $f(x) = -5^x + 7$: $f(0) = -1 + 7 = 6$ ✓. As $x \rightarrow -\infty$: $f(x) \rightarrow 0 + 7 = 7$ ✓.

- A) $f(0) = -1 + 1 = 0 \neq 6$. B) $f(0) = -1 + 5 = 4 \neq 6$. C) $f(0) = -1 + 9 = 8 \neq 6$.
-

Question 9 — Answer: A

Geometry (Surface Area) | Hard

Edge ratio = $7/90$. Surface area ratio = $(7/90)^2 = \mathbf{49/8100}$.

- B) $343/729000$ is the volume ratio. C) $7/540$ uses an incorrect calculation. D) $49/90$ forgets to square.
-

Question 10 — Answer: 55

Geometry (Parallel Lines) | Hard

Triangle MNO: $(4y-5) + y + (3y-23) = 180 \rightarrow 8y - 28 = 180 \rightarrow y = 26$. $M = 99^\circ$, $N = 26^\circ$, $O = 55^\circ$.
 $PQ \parallel MO \rightarrow \text{angle } PQN = \text{angle } MON = \mathbf{55^\circ}$ (corresponding angles, transversal NO).

Question 11 — Answer: A

Systems of Equations | Medium

Adding: $(2x+6) + (2x-6) = (-3y+1) + (3y+27) \rightarrow 4x = 28$. So $\mathbf{4x = 28}$.

- B) 14 is $4x/2$. C) $-12/3 = -4$, unrelated. D) 56 is $2(4x)$.
-

Question 12 — Answer: B

Quadratic Functions | Hard

$p(x) = a(x+1)^2 + 3$. Using $(0, 11)$: $a(1) + 3 = 11 \rightarrow a = 8$. Verify: $p(-3) = 8(4) + 3 = 35 \checkmark$. $p(2) = 8(9) + 3 = 75$. $p(-1) = 3$. $p(2) - p(-1) = 75 - 3 = \mathbf{72}$.

- A) 69 is a computation error. C) 75 is just $p(2)$, not the difference. D) 78 uses an incorrect coefficient.
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Question 13 — Answer: 66

Geometry (Circumscribed Circle) | Hard

Side = $594/3 = 198$. Circumradius $R = 198/\sqrt{3} = 198\sqrt{3}/3 = 66\sqrt{3}$. So $\mathbf{w = 66}$.

Question 14 — Answer: -36

Functions (Evaluation) | Hard

$f(x) = |x|/b + 10$, $b < 0$. $f(16b) = |16b|/b + 10$. Since $b < 0$, $|16b| = -16b$, so $-16b/b = -16$. $f(16b) = -16 + 10 = -6$. $f(4b) = -4 + 10 = 6$. Product: $(-6)(6) = \mathbf{-36}$.

Question 15 — Answer: 4096

Exponential Functions | Hard

$f(x) = cd^{(x/m)}$. $f(6)/f(3) = d^{(3/m)} = 256/16 = 16$. So $d^{(1/m)} = 16^{(1/3)}$. $f(3) = c \cdot (d^{(1/m)})^3 = c \cdot 16 = 16 \rightarrow c = 1$. $f(9) = 1 \cdot (d^{(1/m)})^9 = (16^{(1/3)})^9 = 16^3 = \mathbf{4096}$.

Question 16 — Answer: B

Equivalent Expressions | Easy

$18(x + 7) = 18x + 18(7) = \mathbf{18x + 126}$.

A) Forgot to distribute 18 to 7. C) Added: $18 + 7 = 25$. D) Ignored the 7 entirely.

Question 17 — Answer: D

Ratios | Hard

Original: $50 \text{ in} \times 20 \text{ mi/in} = 1,000 \text{ mi}$ total. Smaller: $50 - 80\%(50) = 10 \text{ in}$. Same area: $1,000 \div 10 = \mathbf{100 \text{ mi/in}}$.

A) $4.00 = 20 \times 0.20$. B) 16.00 applies the reduction to the rate. C) 80.00 is an intermediate error.

Question 18 — Answer: C

Geometry (Similarity) | Hard

Given: $A = D = 65^\circ$. Option C: $B = 50^\circ \rightarrow C = 180 - 65 - 50 = 65^\circ$. $F = 65^\circ \rightarrow E = 180 - 65 - 65 = 50^\circ$. Both triangles have angles $\{65^\circ, 50^\circ, 65^\circ\}$. **Similar by AA.**

A) Same side lengths ($45 = 45$) don't prove similarity when other sides differ. B) Side ratios don't establish SAS without the included angle. D) $B = 65^\circ$, $E = 40^\circ$: $ABC = \{65^\circ, 65^\circ, 50^\circ\}$; $DEF = \{65^\circ, 40^\circ, 75^\circ\}$ — different.

Question 19 — Answer: B

Probability | Medium

$P(\text{bronze}) = 1 - 0.52 - 0.30 = 0.18$. Bronze = $0.18 \times 150 = \mathbf{27}$.

A) $18 = 0.12 \times 150$. C) $45 = 0.30 \times 150$ (silver count). D) $78 = 0.52 \times 150$ (gold count).

Question 20 — Answer: B

Linear Equations (Perpendicular) | Medium

Perpendicular slopes are negative reciprocals. Slope of $j = -1/5 \rightarrow$ slope of $k = 5$.

- A) 9 is the y-intercept. C) $1/5$ is the absolute value of j 's slope. D) $1/9$ confuses y-intercept with reciprocal.
-

Question 21 — Answer: 7

Equivalent Expressions | Easy

$9 + 4 - 6 = 7$. So $9x^3 + 4x^3 - 6x^3 = 7x^3$.

Question 22 — Answer: C

Quadratic Functions | Hard

$q(x) = a(x+2)^2 + 4$. Using $(-1, 13)$: $a(1) + 4 = 13 \rightarrow a = 9$. Verify: $q(0) = 9(4) + 4 = 40$ ✓. $q(1) = 9(9) + 4 = 85$. $q(-2) = 4$. $q(1) - q(-2) = 85 - 4 = 81$.

- A) 72 is a computation error. B) 77 uses an incorrect coefficient. D) 85 is just $q(1)$, not the difference.
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VERIFICATION

All 147 answers have been verified against step-by-step solutions. For any discrepancies, the step-by-step solution above takes precedence over the quick-reference grid.

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